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COAL AND THE EUROPEAN INDUSTRIAL REVOLUTION*

Alan Fernihough and Kevin Hjortshøj O'Rourke

To what extent did growth during the Industrial Revolution depend on coal? We answer the question using a panel of European city sizes between 1300 and 1900. Prior to 1750, there was no relationship between proximity to coalfields and growth; after 1750 cities closer to coalfields grew substantially faster than those further away. We instrument for coal proximity using proximity to Carboniferous-era rock strata.

> We all know the basics of climate change. Human activity has greatly increased the atmospheric stock of greenhouse gases (GHGs), the most important of which, by far, is carbon dioxide (CO_2). . . The additional CO_2 in the atmosphere is due primarily to the massive burning of fossil fuels that has accompanied the industrial revolution.
>
> (Weitzman, 2015, p. 145)

Climate change is refocusing attention on the origins of the fossil fuel economy. Prior to the Industrial Revolution, economies largely relied on wood and charcoal for their energy supplies.[1] In contrast, as early as 1800, when the transition to steam was still in its infancy, English coal production yielded energy that would otherwise have required 11 million acres of woodland to produce, or more than one-third of England's land mass (Wrigley, 1988, pp. 54–5). By the 1820s, British coal production 'liberated' an area as large as the entire island (Sieferle, 2001, p. 103). To what extent did the transition to modern economic growth, which was so positive for long-run living standards, depend on the first of the fossil fuels that are now having such dangerous environmental consequences?

Surprisingly, the extent to which growth during the Industrial Revolution depended on coal remains unclear. On the one hand Wrigley (2010, p. 23) regards the switch to coal as 'a necessary condition for the industrial revolution'. On the other, Mokyr (2009, pp. 101–2) states that 'The Industrial Revolution did not absolutely "need" steam . . . nor was steam power absolutely dependent on coal'. Disagreement on such a central issue in economic history is unsatisfactory, and can only be resolved with comprehensive quantitative evidence.

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The data and codes for this paper are available on the Journal website. They were checked for their ability to reproduce the results presented in the paper.

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[1] Wind, water, animal and human power were the main alternatives.

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In this paper we provide such evidence, using a database of European city sizes spanning the centuries from 1300 to 1900. Controlling for a host of other variables in a difference-in-differences framework, we ask whether cities located closer to coalfields grew faster than those further away. Prior to the middle of the eighteenth century they did not. However, from the late eighteenth century onwards there was a large growth bonus associated with proximity to coal. The effect was large: our preferred specification suggests that cities located 49 km from the nearest coalfield grew 21.1% faster after 1750 than cities located 85 km further away.[2]

A reasonable objection to our argument is that the location of coalfields was endogenous to efforts to find them. We therefore instrument proximity to coalfields with proximity to rock strata from the Carboniferous era. We also look directly at the relationship between city growth and Carboniferous-era rock strata. We control for spatial autocorrelation in a number of ways, estimate using various subsamples of the data, and use a large battery of control variables. Our results are robust across all specifications.

The paper contributes to several literatures, most obviously the debate on the role of coal in the Industrial Revolution. Many classic works in economic history have argued for its importance: Deane (1965), Landes (1966), Braudel (1973), Pollard (1981), Mathias (1983), Wrigley (1988; 2010), Pomeranz (2000), Allen (2009) and Kander *et al.* (2013) to name but a few. On the other side of the debate are authors such as Mokyr (1976; 1983; 2009), McCloskey (1981; 2010) and Clark and Jacks (2007). None of these works provides the kind of quantitative evidence presented here, but a more recent literature uses spatial data of one sort or another, as we do. Crafts and Mulatu (2006), Crafts and Wolf (2014) and Gutberlet (2014) conclude that coal mattered for the location of industry within late nineteenth-century Britain and Germany.[3] In contrast, Wolf (2007) finds no evidence that mineral endowments explained the location of fuel-intensive industries in interwar Poland; Klein and Crafts (2012) find little evidence that coal prices mattered for the location of fuel-intensive industries in the United States between 1880 and 1920; and Martinez-Galarraga (2012) argues that mineral endowments affected the location of mineral-intensive industries in Spain in 1913, but not in other years. Hanlon (2019) finds that by the late nineteenth-century coal-related pollution was lowering city growth in the UK, while Matheis (2016) concludes that the impact of coal production on economic activity in the United States was initially positive but turned negative in the long run. These are all single-country studies: ours covers almost the entire continent of Europe, and uses a data set spanning not only the Industrial Revolution but the centuries preceding it.

The paper also speaks to the voluminous literature on the impact of natural resources on growth (e.g., Wright, 1990; Sachs and Warner, 2001; van der Ploeg, 2011). Our results suggest that the relationship between resources and growth may vary over time, depending on the technological environment. And several papers have used European city sizes to study the determinants of long-run growth as we do here (De Long and Shleifer, 1993; Dittmar, 2011; Bosker *et al.*, 2013; Cantoni, 2015; Hornung, 2015; Squicciarini and Voigtländer, 2015 to name but a few). Our results suggest that geology mattered more for city growth during some historical periods than others.

The following section provides some historical background. Section 2 introduces our data while Section 3 derives the paper's results. Section 4 concludes.

[2] This is a one log point difference in coal proximity evaluated at the median value of 134 km.
[3] Crafts and Wolf find that coal prices mattered, not for the location of cotton mills, but for the size of those mills.

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1. Historical Background

The use of coal did not begin with the Industrial Revolution. It was particularly widespread in Britain, where coal served a wide variety of purposes, both domestic (heating) and industrial: 'brickmaking, glass, ceramics, soapboiling, lime burning, forging, distilling, and brewing' (Mokyr, 2009, p. 22). What changed during the Industrial Revolution was primarily the use of coal in the iron and steel industry and the introduction of the steam engine.

In 1709 Abraham Darby discovered how to smelt iron ore using coke (a purified form of coal) rather than charcoal as a fuel, and the process started becoming widespread in Britain in the second half of the century. In the long run this permitted a vast expansion of the industry. As Wrigley (1988, p. 80) put it, 'Iron . . . has many physical properties that make it of the greatest value to man but as long as the production of 10,000 tons of iron involved the felling of 100,000 acres of woodland, it was inevitable that it was used only where a few hundred-weight or at most a few tons of iron would suffice for the task in hand'. The switch to coal allowed humans to tap into a vast capital reserve of energy—'stored sunlight', as Cipolla (1978, p. 62) calls it—which allowed them to break free of these constraints.

Three years later, in 1712, Thomas Newcomen developed his famous steam engine to pump water from mines. James Watt started working on an improved design in 1763, and by 1776 his steam engines were being used commercially. Steam then started to diffuse across the economy, slowly at first, and then more rapidly, so that eventually coal was being used to power not just the metallurgical industries, but textiles and many other sectors as well. Steam accounted for 35,000 out of the 170,000 horsepower installed in Britain in 1800; for 165,000 out of 350,000 in 1830; for 2,060,000 out of 2,300,000 in 1870; and for 9,659,000 out of 9,842,000 in 1907 (Crafts, 2004, p. 342). The new coal-using technologies diffused with a lag to the rest of Europe. By the middle of the 19th century, both coke-smelting and steam engines were being used in all the coalfields of northern France, Belgium and western Germany (Wrigley, 1961, p. 4).

Coal was bulky, heavy and costly to transport. It was also a fuel, whose weight vanished when it was used in the production process: there were thus substantial cost savings if coal was used close to where it was mined (Wrigley, 1961, pp. 6–7). According to Mathias (1983, p. 11), 'The logistics of energy inputs based upon coal, translated against available transport in a pre-railway age, precluded any major industrial complex in heavy industry from developing except where coal and ore were plentiful and adjacent to one another or to water carriage'. Pollard (1981, p. 4) remarks that 'the map of the British Industrial Revolution, it is well known, is simply the map of the coalfields'. On the European continent, the coalfields of Belgium and northern France, and later on the Ruhr, became major centres of heavy industry, and other industrial regions 'only survived if they had reasonable access by water to a supply of good coal' (Pollard, 1981, p. 121). Countries such as Italy that lacked domestic coal supplies were severely handicapped relative to their more fortunate counterparts (Bardini, 1997).

Not all economic historians agree that proximity to coalfields was essential for the development of industry. Mokyr (1976, pp. 204–8) argues that local supplies of coal in Belgium cannot explain why it industrialised, while the Netherlands did not: the Dutch could import coal by sea, and use both peat and wind. In a similar vein, he dismisses the argument that pre-Famine Ireland did not industrialise because of a lack of suitable coal deposits (Mokyr, 1983, pp. 152–8). Ireland imported coal from Britain, with the result that its coal prices were between 100% and 150% higher; fuel costs in the 'nonmetallurgical industries' were at most 4% of total costs. The lack of suitable local coal supplies thus increased Irish costs by at most 10% relative to British costs,

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and by less once substitution possibilities are taken into account. Lower Irish wages, he argues, should have more than compensated for this.

If the introduction of new coal-using technologies meant that growth became more coal-dependent during the Industrial Revolution then you would expect that coal would have started to matter for city growth at some point in the late eighteenth century or nineteenth century. Alternatively, given that coal was used for a variety of purposes prior to the eighteenth century, proximity to it may have mattered for city growth even earlier. And a third possibility is that it never really mattered at all. In the following section we outline the data that we will use to distinguish between these alternative hypotheses.

2. Data

Our empirical analysis combines a number of different sources. To measure economic activity we use population size for a panel of European cities based in the first instance on Bairoch *et al.* (1988). The panel consists of 2,180 cities that satisfy the criterion of having at least 5,000 inhabitants at some stage between 800 and 1850. City populations are measured at 100-year intervals between 800 and 1700, and at 50-year intervals after 1700. The panel is unbalanced, since evidence on city population levels is understandably lacking for many cities as we look further back in time.

These missing observations are potentially problematic, and we therefore take a number of steps to remedy this. We begin our analysis in 1300, ignoring all data prior to this date. The vast majority of cities included in the Bairoch *et al.* data set do not have population data prior to 1300, and we also suspect that the available population evidence prior to 1300 is less accurate than more recent figures. While our baseline regressions are performed on the unbalanced sample excluding missing data we also replicate our analysis using a balanced panel in which missing city population sizes were interpolated. The results of this exercise show that excluding missing city size data does not make any material difference to our results.[4]

Bairoch *et al.* (1988) provide city population sizes up until 1850. Given that we want to measure economic development through to the end of the nineteenth century, we extend the Bairoch *et al.* panel to include city populations in 1900. For this purpose we use a contemporary resource: *Lippincott's New Gazetteer* (Heilprin and Heilprin, 1906). The gazetteer lists location and population for the majority of the cities included in the Bairoch *et al.* panel. We restrict our sample to cities positioned within the −20 and 50 degree lines of longitude. This leaves a sample of 2,180 cities.

The aim of this paper is to link city growth to the availability of coal. Thus, we need to create a measure of access to coal for each city. To do this we digitise the *Les Houillères Européennes* map in Châtel and Dollfus (1931). This atlas shows the location of 124 major coalfields within Europe. We include lignite fields in our calculations since lignite (or brown coal) played an important role in the introduction of the steam engine in Prussia (Redlich, 1944). We digitise this map, and then calculate the minimum great-circle distance from each city i to the nearest coalfield. Figure 1 illustrates the location of our cities and their distance to coalfields. Because it is more intuitive to interpret the effect of proximity to coal than the effect of distance from it (so

[4] See Online Appendix C. We interpolated city population sizes using a simple algorithm that modelled city population size, in natural logarithms, as a linear function of time. In cases where the predicted city size was below 500 we bottom coded at 500. We found that this approach worked best in practice, and indeed better than more technically sophisticated methods. The in-sample correlation of logged population and its fit based on the aforementioned method was 0.937.

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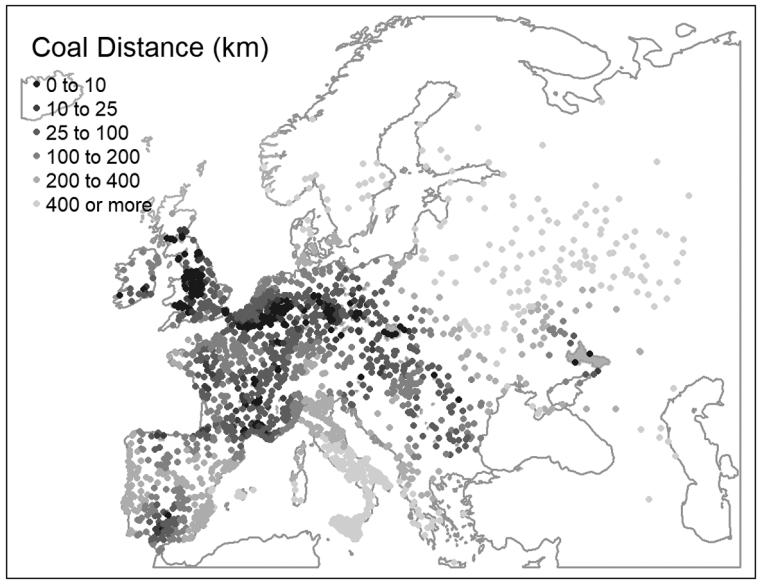


Fig. 1. *Cities' Distance to Coalfields.*

Source. See text.

there is a positive relationship if coal drives city population growth), we define the coal variable $Coal_i$ used in our regressions by inverting the distance from each city to the nearest coalfield.[5]

Coal prices, as used in Crafts and Wolf (2014), would be an economically more meaningful measure of access to coal. Unfortunately, we lack city-level price data for our panel of more than 2,000 cities over 600 years. However, we assume that our proximity measure is highly correlated with the spatial variation in coal prices. To investigate how reasonable this assumption is, we use data from Crafts and Wolf (2014), who provide coal prices for all British and Irish counties between 1843 and 1845. We match each of the British and Irish cities in our sample to their county, and allocate that county's price to them. We then compare the coal price for each city with the proximity of that city to the closest coalfield. This comparison is illustrated in Figure 2. The strength of the relationship between these two measures is evident (the correlation coefficient is −0.742): coal prices were lower in cities closer to coalfields. Such a strong correlation is striking in a comparatively small country like the UK, well-endowed as it was with both coalfields, internal waterways and a navigable coastline: our assumption that proximity to coalfields was negatively correlated with coal prices seems a valid one.

In some specifications, we instrument our proximity to coal measure with a variable that measures the proximity of cities to Carboniferous-era geological strata. Coal is often found in rock strata from the Carboniferous age, and thus the coalfield locations should often be on, or very close to, rock strata from the Carboniferous epoch.[6] To construct a Carboniferous measure that corresponds to our coal proximity variable, we use data collated by the German Federal Institute for Geosciences and Natural Resources (BGR) for a project that mapped the European geological

[5] Cities located on coalfields are assigned an arbitrary distance of 1 km. Appendix A details how all variables used in the regressions were constructed.

[6] Carboniferous means 'coal-bearing'.

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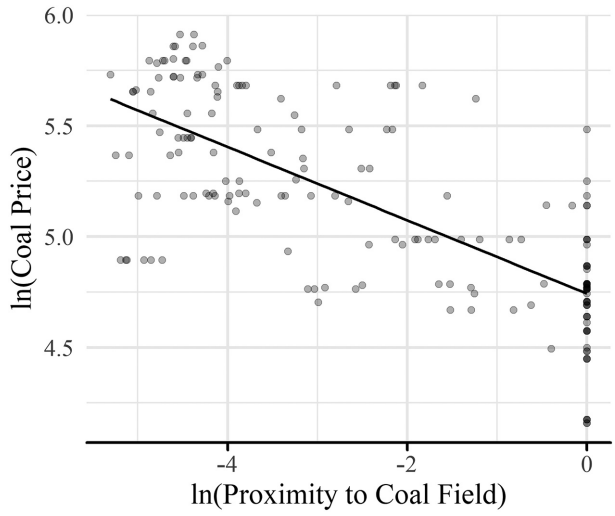


Fig. 2. *Proximity to Coalfields and Coal Prices, United Kingdom 1843–5.*
Source. See text.

landscape: the 1:5 Million International Geological Map of Europe and Adjacent Areas (Asch, 2005). This was a pan-European project that involved a high level of collaboration across several national geological offices; the result was a high-resolution GIS map containing a number of geological features including age and rock type. We create a proximity to Carboniferous strata variable for each city in an equivalent manner to that of the coal proximity variable.

Most coalfields overlap with areas whose rock strata are categorised as being of the Carboniferous epoch, although this overlap is not perfect. There are some Carboniferous areas that do not contain any coalfields, and some coalfields not located within a Carboniferous area. For example, lignite is geographically younger than black coal, typically originating in the Tertiary period. Our coal and Carboniferous measures are correlated, but not perfectly. Figure 3 illustrates the strength of this relationship.[7]

3. Empirical Analysis

Figure 4 previews our main result. We divide cities into two groups: those within 25 km of a coalfield, and those further away. For each group, we plot the population from 1300 to 1900, indexed at 1750.[8] As can be seen, the two groups of cities grew at similar rates prior to 1750. After 1750 cities located within 25 km of a coalfield grew roughly twice as fast as those further away: by 74% as opposed to 33% in the late eighteenth century, by 111% as opposed to 57% in the early nineteenth, by 123% as opposed to 88% in the late nineteenth, and by 721% as opposed

[7] Both variables are truncated at zero, the point at which cities lie directly on either coalfields or Carboniferous strata.
[8] We present population levels indexed at 1750 due to our sample being unbalanced. Population levels for each period are calculated by applying population growth rates forwards and back in time from 1750. Population growth rates are calculated for centuries (prior to 1700) or half-centuries (after 1700) using the entire (unbalanced) sample of city sizes; we use unweighted average growth rates.

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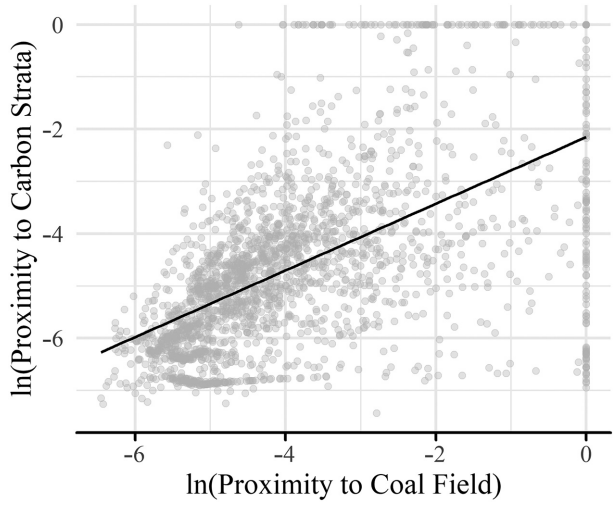


Fig. 3. *Bivariate Relationship between Proximity to Coal and Carboniferous Rock Strata. Source.* See text.

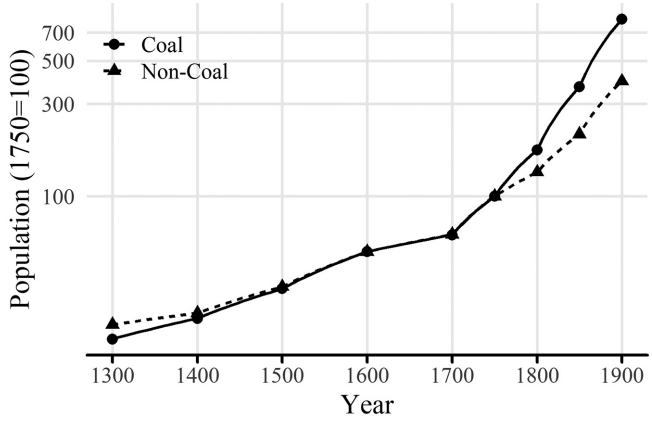


Fig. 4. *Population in Cities Located More and Less than 25 km from Coalfields, 1300–1900. Source.* See text.

to 293% over the 150 years as a whole. We obtain similar results when we expand or contract the 25 km threshold, or include interpolated population sizes.[9]

Figure 5 repeats the analysis for three groups of countries: the United Kingdom, where the new coal-using technologies first originated; Belgium, France and Germany, the earliest imitators; and the rest of Europe. As can be seen, in the UK, proximity to coal became associated with rapid growth in the late eighteenth century; in Belgium, France and Germany the association emerged in the early nineteenth century; while elsewhere it only became discernible in the late nineteenth century. The raw data thus support the accounts of technological diffusion given in

[9] See Online Appendix B.

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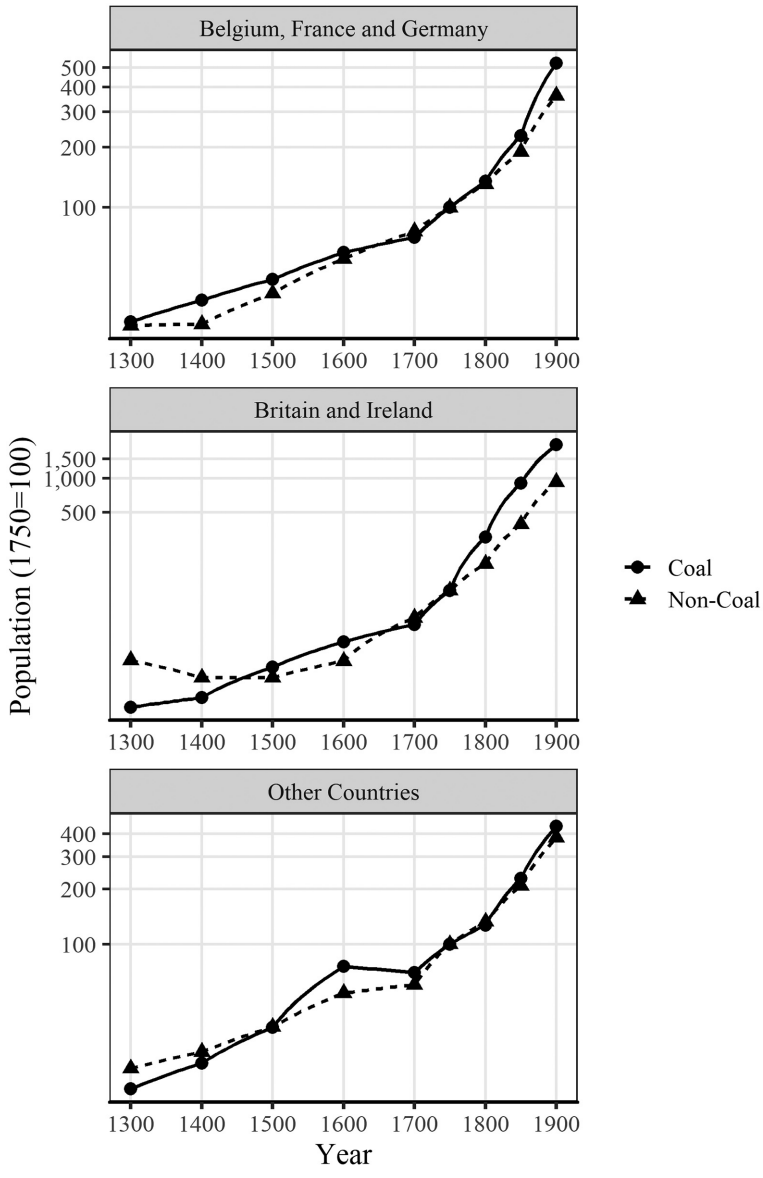


Fig. 5. *Population in Cities Located More and Less than 25 km from Coalfields, 1300–1900. Source.* See text.

such standard sources as Tann and Breckin (1978), and suggest that proximity to coal became relevant for economic growth once the technological environment had changed.

The evidence in Figures 4 and 5 is persuasive but not conclusive. Access to coal was not a dichotomous variable, but something that presumably varied continuously with distance. In the regressions we therefore use our great-circle distance-based proximity measure, but this could bias our results for a number of reasons. As noted earlier we are using proximity to coal rather

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than coal prices. Furthermore, we are not using a least-cost measure of coal proximity, like Özak (2018), since dramatic changes in the form and speed of transportation methods over our six century sample period make such a calculation impracticable. Another issue is that while we can measure the distance to each coalfield, we cannot measure the abundance or quality of the coalfield. Our main explanatory variable is therefore measured with error. In addition, we cannot rule out the possibility of reverse causality, that is to say the possibility that the coalfields in our data set were developed because of their proximity to cities. As noted earlier we deal with these problems by using proximity to Carboniferous-era rock strata as an instrument for proximity to coal.

Omitted variable bias poses a further threat to our identification strategy. We therefore include a rich set of control variables in our regression analysis. These include three variables that measure the distance to the Atlantic, the Mediterranean and to all other coasts, as well as cities' distances to primary and secondary rivers. We also follow the literature in including variables measuring terrain ruggedness (Nunn and Puga, 2010); the suitability of land for cultivating potatoes, wheat and oats (Nunn and Qian, 2011); altitude; and temperature (median and range). We include controls for absolute latitude and longitude in all model specifications. One would not expect any of these factors to have the same impact on growth across six centuries, and so in all cases we interact our control variables with time period dummies. In addition, we include city fixed effects to control for any time-invariant determinants of individual cities' growth. Appendix A provides information on the construction of these variables and the sources used.

We use a conventional difference-in-differences (DID) model, to see whether cities that were located closer to coalfields grew more rapidly during and after the Industrial Revolution (but not before) than those further away. In particular, we estimate city population as a function of the interaction between a city's proximity to coal and a post-treatment year indicator. The 'treatment' in this case is the introduction of the new coal-using technologies associated with the Industrial Revolution, and based on the evidence in Figure 4, as well as the historical record, we take this to have occurred after 1750. We therefore estimate:

$$\ln(Pop_{it}) = \alpha_i + \gamma_t + \beta \ln(Coal_i)\, I_t^{Post} + \sum_{j=1400}^{1900} \mathbf{X_i'} I_t^j \Psi_{\mathbf{j}} + \varepsilon_{it}, \quad (1)$$

where Pop_{it}, the population of city i in year t, is a function of city (α_i) and year (γ_t) fixed effects, an interaction between a dichotomous indicator for post-1750 observations (I_t^{Post}) and the natural logarithm of our coal proximity measure $Coal_i$, and finally our full set of control variables $\mathbf{X_i'}$ interacted with year fixed effects I_t^j. We are particularly interested in β, which measures the relationship between coal proximity and post-1750 city growth.

Table 1 details our main regression results. Column (1) displays the OLS estimate of β for our unbalanced panel of European cities from 1300 to 1900. The reported coefficient of 0.108 indicates that cities that were closer to coal grew faster after 1750: a 10% increase in coal proximity resulted in a 1% increase in city growth. Alternatively, a city located 49 km from a coalfield grew 10.8% more rapidly than a city located 85 km (a one log point difference) further away.[10] This relationship is conditional on our full set of time-varying control variables discussed above and detailed in Appendix A. The reported standard errors have been adjusted to account for spatial (Conley, 1999) and temporal (Newey and West, 1994) autocorrelation.

[10] The latter distance, 134 km, was the median distance from a coalfield in our sample.

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Table 1. *Logged Population on Coal/Carbon, OLS and IV.*

	Dependent variable:			
	ln(Population)		ln(Coal)×Post1750	ln(Population)
	(1)	(2)	(3)	(4)
ln(Coal)×Post1750	0.108***			0.211***
	(0.021)			(0.034)
ln(Carbon)×Post1750		0.085***	0.404***	
		(0.018)	(0.053)	
Method	OLS	Reduced-Form OLS	2SLS, First Stage	2SLS, Second Stage
F-Test Statistic			58.980	
Base Controls	Y	Y	Y	Y
Observations	11,161	11,161	11,161	11,161

Notes: Column (1) reports the β estimate resulting from an OLS regression of logged city population size on an interaction term composed of logged coal proximity times a post-1750 dummy. Column (2) reports the results from a reduced form regression of logged city population on the interaction between a post-1750 dummy and the logged proximity to Carboniferous rock strata. Columns (3) and (4) display the first and second stage regressions of a 2SLS regression of logged city population size on the logged coal proximity–post-1750 interaction, where the coal variable is instrumented using a logged Carboniferous strata proximity–post-1750 interaction. All regressions include our set of control variables. Standard errors adjusted to account for spatial and temporal autocorrelation using Conley (uniform kernel radius of 500 km) and Newey–West (Bartlett Kernel of 500 years) corrections. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

The association between coal and post-1750 population growth is apparent in column (1). To what extent is the relationship causal? To tackle this question we introduce our proximity to Carboniferous strata measure into the analysis. Column (2) models the conditional reduced-form relationship between Carboniferous-strata proximity and city population size, specified in identical manner to the OLS relationship between coal and city size in column (1). It demonstrates that Carboniferous strata were related to city growth in post-eighteenth-century Europe, with estimated coefficients similar in magnitude to those in column (1).

Next we model coal proximity's influence on city growth using proximity to Carboniferous strata as an instrument and two-stage least squares (2SLS). The first-stage regression results in column (3) show, unsurprisingly, a strong relationship between Carboniferous strata and coal proximity. Reassuringly, the first-stage *F*-test statistic (adjusted for spatial and temporal autocorrelation as per the standard errors) on the excluded instrument is 58.98. The second-stage regression results reported in column (4) confirm that coal proximity underpinned urban population growth during the Industrial Revolution. Indeed, the estimate is about twice as large (0.211) as the corresponding OLS estimate. The impact of a one log point change in coal proximity at the median (i.e., comparing cities located 134 km and 49 km from the nearest coalfield respectively) is a 21.1% difference in population growth.

Table 1 points towards a large and statistically robust effect of coal access on urban growth after 1750. The results also indicate that the OLS estimate is biased downwards. Could classical measurement error in our coal variable account for this? We can quantitatively assess this possibility by returning to Figure 2. If we use coal proximity in our regressions instead of coal prices (which are what should really have mattered economically, but which we do not have for continental Europe) we can assume the following relationship:

$$Coal_i = -Price_i + v_i. \tag{2}$$

In (2) a *Z*-score standardised coal proximity variable is taken to be a one-for-one function of *Z*-score transformed coal prices and random measurement error v_i. The variance of v_i determines

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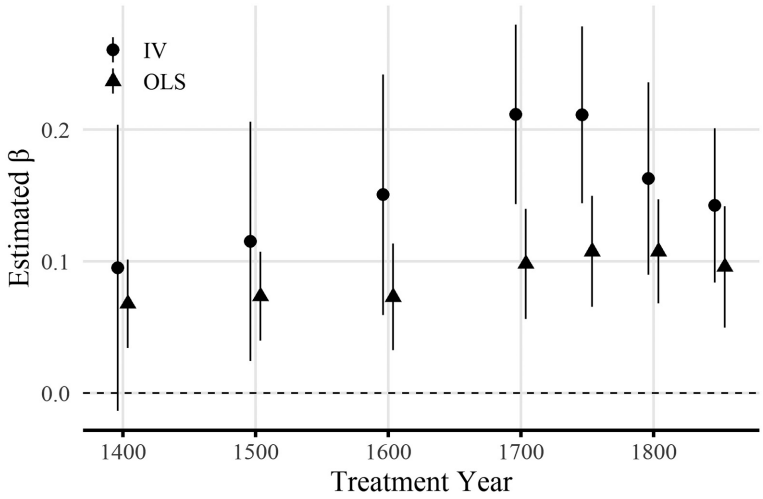


Fig. 6. *Estimated Coal Effect, Alternative Treatment Dates.*
Notes: The figure reports the β estimates resulting from OLS and second-stage 2SLS regressions of logged city population size on logged coal proximity interacted with dummies representing alternative treatment dates. The IV estimates instrument the coal interaction variables using the appropriate post-treatment logged Carboniferous strata interaction variables. All regressions include our control variables and city and year fixed effects. Standard errors adjusted to account for spatial and temporal autocorrelation using Conley (uniform kernel radius of 500 km) and Newey–West (Bartlett Kernel of 500 years) corrections.
Source. See text.

the size of the potential bias via the signal-to-noise ratio: $\frac{\sigma^2_{Price}}{\sigma^2_{Price}+\sigma^2_v} = \frac{1}{1+\sigma^2_v}$, since we know that the variance of the standardised price is unity. Our estimate of $\hat{\sigma}^2_v$ from the British and Irish data is 0.569, implying that measurement error could deflate a coal price coefficient by a factor of 0.637. This simple exercise suggests that measurement error can indeed potentially explain why our preferred instrumental variables estimate of the coal effect is larger than its single equation OLS counterpart.

If the correlation between proximity to coal and city growth that we have uncovered reflects the introduction of new coal-using technologies, and if the most appropriate treatment date is the middle of the eighteenth century, as the qualitative literature on the Industrial Revolution suggests, then our estimated coal effect should be diluted if we use less appropriate treatment dates, becoming smaller if we move either forwards or backwards in time from 1750. Figure 6 plots the β-coefficients (and associated 95% confidence intervals) obtained when we estimate equation (1) using alternative treatment dates. As can be seen, the coefficients decline monotonically as we move away from our preferred 1750 treatment date, and the IV estimate becomes statistically insignificant if we use the earliest possible treatment date, namely 1400. Proximity to coal did not matter uniformly across time: as many traditional economic historians have argued, and as Figure 4 suggested, it became important as a result of the Industrial Revolution.

We performed a battery of robustness checks, the results of which (in the interests of parsimony) are presented in Online Appendix C. Our first robustness exercise examines different control specifications. Table 1 included a large set of time-varying control variables. Online Appendix Table C.1 shows that the estimated coal effect is qualitatively similar if you either

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remove these time-varying controls, or additionally include time-varying state border fixed effects that control for institutions, public health infrastructure, broader demographic change, and a host of other factors that may influence city population size at a national, rather than a subnational, level in different time periods. Online Appendix Table C.2 shows that our results are robust to excluding British and Irish cities, although the coefficients are slightly smaller: our findings in Table 1 are not an artefact of the UK's industrial exceptionalism. Online Appendix Table C.3 explores the potential importance of missing city size data by using balanced (interpolated) panel data. Once again our results survive. Online Appendix Table C.4 shows that our results are robust to removing cities closer than 10 km, 25 km and 50 km to coalfields, and Online Appendix Table C.5 shows that they are also robust to using the methods described in Kelly (2020) to deal with spatial autocorrelation.

4. Conclusion

The Industrial Revolution was above all a technological phenomenon. However, the nature of the new technologies meant that access to coal became an important driver of regional economic development in the nineteenth century. We have found a clear relationship between proximity to coal and city growth after 1750, consistent with the accounts of Wrigley (2010) and many other traditional economic historians: geology mattered for economic growth during this period. While innovation may have been the ultimate source of economic development, we cannot ignore the role of fossil fuels in fuelling growth after the Industrial Revolution, or of geography in determining who experienced that growth over the course of the nineteenth century.

Appendix A. Data Sources

City Population Size. Sources: Heilprin and Heilprin (1906); Bairoch *et al.* (1988). Data partially retrieved from: https://github.com/JakeRuss/bairoch-1988.

Coal. Minimum distance (km) from any of Europe's major coalfields. When a city is positioned within a coalfield we assume an arbitrary distance of 1 km. We transform this value into a proximity measure by dividing into one (inverse-distance measure). Source: Châtel and Dollfus (1931).

Carbon. Minimum distance (km) from any onshore geological area classified as being Carboniferous in the IGME 5000. When a city is positioned within such an area we assume an arbitrary distance of 1 km. We transform this value into a proximity measure by dividing into one (inverse-distance measure). Source: Asch (2005).

Latitude and Longitude. The line of absolute latitude/longitude a city is positioned on. We use these data to construct both the Coal and the Carbon variables. Source: Bairoch *et al.* (1988).

Potato, Wheat, and Oats. Suitability of city location for growing specific crops. Data were constructed as in Nunn and Qian (2011), measuring the amount of land suitable for cultivating each particular crop within 100 km of the city location. The underlying data for these variables originate from a series of raster images produced by the Food and Agriculture Organization of the United Nations (FAO) under the Global Agro-Ecological Zones (GAEZ) assessment methodology. Our data come from the data portal released in 2011. We use the crop suitability index class to define whether or not land is suitable for cultivating each crop. Like Nunn and Qian, we consider land suitability defined as very high, high, good, and medium as suitable, and

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land in other classes as unsuitable. We use rain-fed crop conditions with an intermediate input level, to capture historical conditions as accurately as possible. The raster image that underpins these data is in a 5 arc-minute resolution, meaning that the raster cells are typically (although this changes based on distance to the equator) 10 km apart. Our suitability variable measures the area of raster cells considered suitable for growing the crop in question where the centroid of the raster cell lies within a 100 km radius of the city location. This yields a measure in squared km. We take the natural logarithm of this value, after adding the arbitrary value of 1 as some cities have no land suitable for a particular crop. Sources: FAO/IIASA (2011). Data retrieved from: http://gaez.fao.org/Main.html#.

Altitude. Values were extracted from a raster image produced by the FAO, and relate to the median meters above sea level measured within each raster cell. We take the natural logarithm of this value, after adding the arbitrary value of 26 m as some cities are slightly below sea level. Sources: FAO/IIASA (2011). Data retrieved from: http://www.fao.org/nr/gaez/en/.

Ruggedness. Average terrain ruggedness within 25 km of the city location. The underlying data for this variable originate from the altitude raster image produced by the FAO. Our data come from the data portal released in 2011. We convert this altitude raster image to terrain ruggedness indices using the method proposed in Wilson *et al.* (2007), where each cell represents the mean of the absolute differences between the value of that cell and the value of its 8 surrounding cells. The raster image that underpins these data is in a 5 arc-minute resolution, meaning that the raster cells are typically (although this changes based on distance to the equator) 10 km apart. We take the natural logarithm of this value after adding the arbitrary value of 1. Sources: FAO/IIASA (2011). Data retrieved from: http://gaez.fao.org/Main.html#.

Temperature. The mean annual temperature for each city. The underlying data were extracted from a GIS raster image showing the mean annual temperature calculated over the period 1961–1990. We then take the natural logarithm of this value. Sources: FAO/IIASA (2011). Data retrieved from: http://gaez.fao.org/Main.html#.

Atlantic, Mediterranean, Coast, and Rivers. We use 1:10 m Physical Vector data on coastlines and rivers and measure the minimum distance from each city to each of these individual features. The minimum distance to the coast relates to all coasts worldwide. We separate the Atlantic from the Baltic and Mediterranean seas at the most northerly point of Denmark and most southerly point of Spain respectively. We then transform these measures into proximity values by inverting, before taking the natural logarithm to normalise the distribution. Data retrieved from: http://bit.ly/1caWOLk.

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Additional Supporting Information may be found in the online version of this article:

Online Appendix
Replication Package

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